

A Pair-Copula Families

This Section of the Appendix contains the closed form copula densities and h-functions of the bivariate families used in our implementation. Each family is parametrized by a parameter vector θ whose admissible value ranges are reported in Table A1. For a bivariate copula $C(u_1, u_2; \theta)$ the copula density is defined as:

$$c(u_1, u_2; \theta) = \frac{\partial^2 C(u_1, u_2; \theta)}{\partial u_1 \partial u_2},$$

while the h-functions correspond to the conditional distributions:

$$h_1(u_1, u_2; \theta) = \frac{\partial C(u_1, u_2; \theta)}{\partial u_2}, \quad h_2(u_1, u_2; \theta) = \frac{\partial C(u_1, u_2; \theta)}{\partial u_1}.$$

Since $h_2(u_1, u_2; \theta) = h_1(u_2, u_1; \theta)$ for all families considered, we report only the first h-function h_1 .

Family	1st Parameter	2nd Parameter	Lower TD	Upper TD
Gaussian	$\rho \in [-1, 1]$	—	0	0
Student-t	$\rho \in [-1, 1]$	$\nu \in [2, 30]$	+	+
Clayton	$\delta \in [0, 7.5]$	—	+	0
Frank	$\theta \in [-10, 10] \setminus 0$	—	0	0
Gumbel	$\delta \in [1, 17]$	—	0	+
Joe	$\delta \in [1, 10]$	—	0	+

Table A1. Implemented bivariate copula families with their parameter ranges and tail dependence behavior. + indicates non-zero tail dependence; 0 indicates tail independence.

A.1 Gaussian Copula

The Gaussian copula is parametrized by a single parameter, the correlation coefficient ρ . It admits values in $\rho \in (-1, 1)$, with $\rho = 0$ denoting independence between the two variables. Letting $z_1 = \Phi^{-1}(u_1)$ and $z_2 = \Phi^{-1}(u_2)$, where Φ denotes the standard normal CDF, the copula density of the Gaussian copula is:

$$c(u_1, u_2; \rho) = \frac{1}{\sqrt{1 - \rho^2}} \exp \left(-\frac{\rho^2 (z_1^2 + z_2^2) - 2\rho z_1 z_2}{2(1 - \rho^2)} \right)$$

and the h-function is:

$$h_1(u_1, u_2; \rho) = \Phi \left(\frac{z_1 - \rho z_2}{\sqrt{1 - \rho^2}} \right).$$

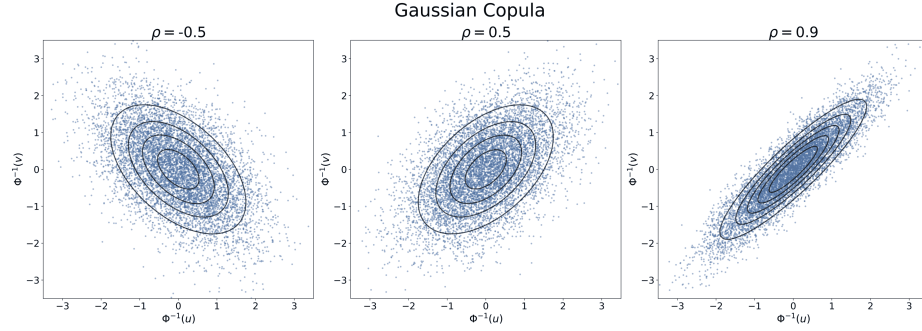


Fig. A1. Joint densities induced by the Gaussian copula for different values of ρ . All panels use the same scale to ensure comparability.

Fig. A1 illustrates the dependence structures induced by the Gaussian copula for different values of the correlation coefficient ρ . The displayed joint densities are obtained by combining the copula density for the specific parameter configuration with standard Gaussian marginals and are overlaid on samples simulated from the corresponding copula model.

A.2 Student-t Copula

The Student-t copula is parametrized by two parameters, the correlation coefficient ρ and the degrees of freedom ν . They admit values in $\rho \in (-1, 1)$ and $\nu > 0$, with smaller values of ν corresponding to heavier tails and stronger tail dependence, while larger values induce a behavior closer to a Gaussian copula. Letting $z_1 = t_\nu^{-1}(u_1)$ and $z_2 = t_\nu^{-1}(u_2)$, where t_ν denotes the standard Student-t CDF with ν degrees of freedom, the copula density is:

$$c(u_1, u_2; \rho, \nu) = \frac{\Gamma(\frac{\nu+2}{2}) \Gamma(\frac{\nu}{2})}{\sqrt{1-\rho^2} \Gamma(\frac{\nu+1}{2})^2} \cdot \frac{\left(1 + \frac{z_1^2 + z_2^2 - 2\rho z_1 z_2}{\nu(1-\rho^2)}\right)^{-\frac{\nu+2}{2}}}{\left(1 + \frac{z_1^2}{\nu}\right)^{-\frac{\nu+1}{2}} \left(1 + \frac{z_2^2}{\nu}\right)^{-\frac{\nu+1}{2}}}$$

and the h-function is:

$$h_1(u_1, u_2; \rho, \nu) = t_{\nu+1} \left(\frac{z_1 - \rho z_2}{\sqrt{\frac{(\nu + z_2^2)(1-\rho^2)}{\nu+1}}} \right).$$

Fig. A2 illustrates the dependence structures induced by the Student-t copula for different values of the parameters ρ and ν . The displayed joint densities are obtained by combining the copula density for the specific parameter configurations with standard Gaussian marginals and are overlaid on samples simulated from the corresponding copula model.

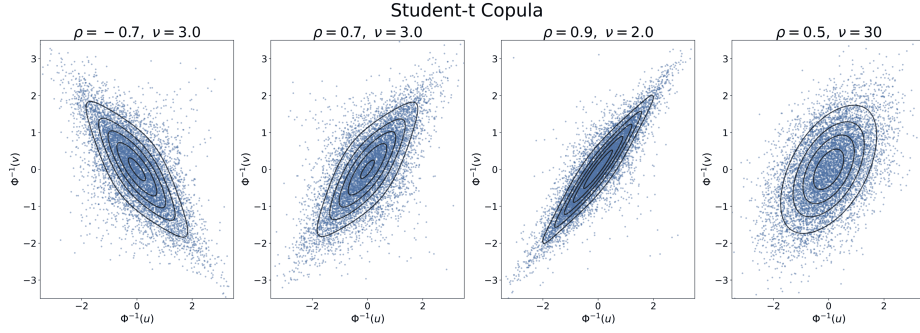


Fig. A2. Joint densities induced by the Student-t copula for different values of ρ and ν . All panels use the same scale to ensure comparability.

A.3 Clayton Copula

The Clayton copula is parametrized by a single parameter δ controlling the strength of dependence, which admits values in $\delta > 0$, with larger values corresponding to stronger lower-tail dependence. The copula density is:

$$c(u_1, u_2; \delta) = (1 + \delta) (u_1 u_2)^{-1-\delta} (u_1^{-\delta} + u_2^{-\delta} - 1)^{-1/\delta-2},$$

and the h-function is:

$$h_1(u_1, u_2; \delta) = u_2^{-\delta-1} (u_1^{-\delta} + u_2^{-\delta} - 1)^{-1-1/\delta}.$$

Fig. A3 illustrates the dependence structures induced by the Clayton copula for different values of the parameter δ . The displayed joint densities are obtained by combining the copula density for the specific parameter configurations with standard Gaussian marginals and are overlaid on samples simulated from the corresponding copula model.

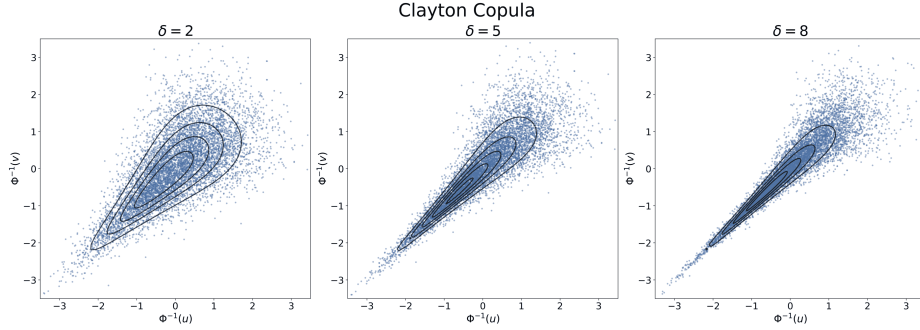


Fig. A3. Joint densities induced by the Clayton copula for different values of δ . All panels use the same scale to ensure comparability.

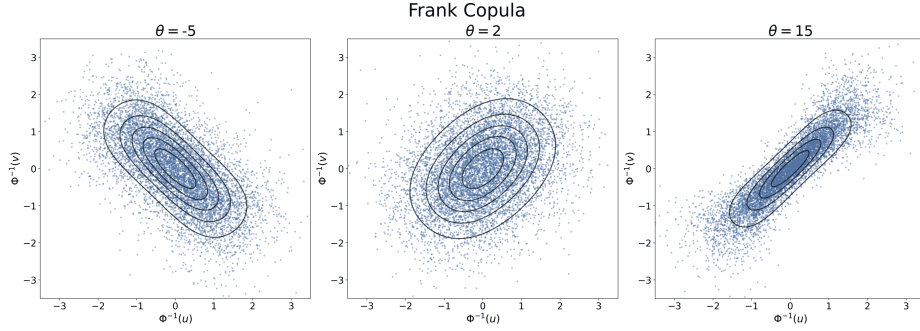


Fig. A4. Joint densities induced by the Frank copula for different values of θ . All panels use the same scale to ensure comparability.

A.4 Frank Copula

The Frank copula is parametrized by a single parameter θ , controlling the strength and direction of dependence, which admits values in $\theta \neq 0$; positive values correspond to positive dependence and negative values to negative dependence, with the independence copula recovered as $\theta \rightarrow 0$. The copula density is:

$$c(u_1, u_2; \theta) = \frac{\theta (1 - e^{-\theta}) e^{-\theta(u_1 + u_2)}}{[(1 - e^{-\theta}) - (1 - e^{-\theta u_1})(1 - e^{-\theta u_2})]^2},$$

and the h-function is:

$$h_1(u_1, u_2; \theta) = \frac{e^{-\theta u_2} (1 - e^{-\theta u_1})}{e^{-\theta u_1} + e^{-\theta u_2} - e^{-\theta(u_1 + u_2)} - e^{-\theta}}.$$

Fig. A4 illustrates the dependence structures induced by the Frank copula for different values of the parameter θ . The displayed joint densities are obtained by combining the copula density for the specific parameter configurations with standard Gaussian marginals and are overlaid on samples simulated from the corresponding copula model.

A.5 Gumbel Copula

The Gumbel copula is parametrized by a single parameter δ , controlling the strength of dependence, which admits values in $\delta \geq 1$, with larger values corresponding to stronger upper-tail dependence and $\delta = 1$ recovering independence.

With $C(u_1, u_2; \delta) = \exp\left[-\{(-\log u_1)^\delta + (-\log u_2)^\delta\}^{1/\delta}\right]$ the density is:

$$\begin{aligned} c(u_1, u_2; \delta) &= C(u_1, u_2; \delta) \cdot (u_1 u_2)^{-1} \\ &\times \{(-\log u_1)^\delta + (-\log u_2)^\delta\}^{1/\delta - 2} \\ &\times [(-\log u_1)(-\log u_2)]^{\delta - 1} \\ &\times \left[\{(-\log u_1)^\delta + (-\log u_2)^\delta\}^{1/\delta} + \delta - 1\right], \end{aligned}$$

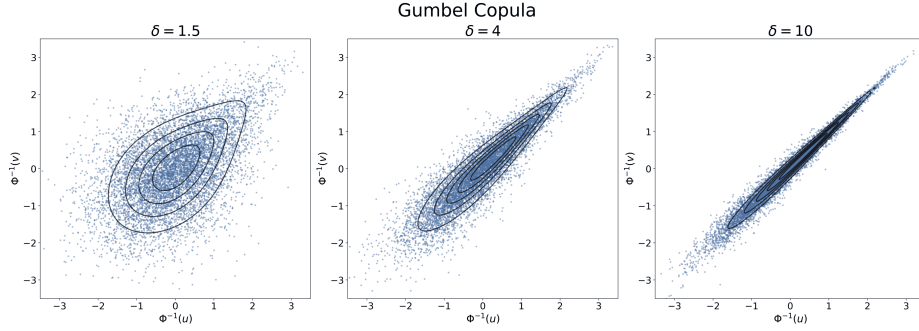


Fig. A5. Joint densities induced by the Gumbel copula for different values of δ . All panels use the same scale to ensure comparability.

and the h-function is:

$$h_1(u_1, u_2; \delta) = C(u_1, u_2; \delta) \cdot \frac{1}{u_2} \cdot (-\log u_2)^{\delta-1} \cdot \{(-\log u_1)^\delta + (-\log u_2)^\delta\}^{1/\delta-1}.$$

Fig. A5 illustrates the dependence structures induced by the Gumbel copula for different values of the parameter δ . The displayed joint densities are obtained by combining the copula density for the specific parameter configurations with standard Gaussian marginals and are overlaid on samples simulated from the corresponding copula model.

A.6 Joe Copula

The Joe copula is parametrized by a single parameter δ , controlling the strength of the dependence, which admits values in $\delta \geq 1$. Larger values correspond to stronger upper-tail dependence. With $t_1 = (1 - u_1)^\delta$ and $t_2 = (1 - u_2)^\delta$ the copula density can be expressed as:

$$\begin{aligned} c(u_1, u_2; \delta) &= (t_1 + t_2 - t_1 t_2)^{1/\delta-2} \\ &\quad \times (\delta - 1 + t_1 + t_2 - t_1 t_2) \\ &\quad \times [(1 - u_1)(1 - u_2)]^{\delta-1}, \end{aligned}$$

and the h-function is:

$$h_1(u_1, u_2; \delta) = (1 - t_1) \cdot (1 - u_2)^{\delta-1} \cdot (t_1 + t_2 - t_1 t_2)^{1/\delta-1}.$$

Fig. A6 illustrates the dependence structures induced by the Joe copula for different values of the parameter δ . The displayed joint densities are obtained by combining the copula density for the specific parameter configurations with standard Gaussian marginals and are overlaid on samples simulated from the corresponding copula model.

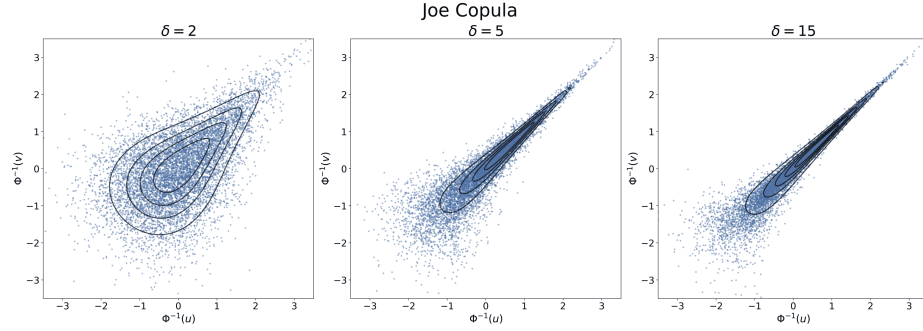


Fig. A6. Joint densities induced by the Joe copula for different values of δ . All panels use the same scale to ensure comparability.

A.7 Tail Dependence Coefficients

The dependence structure captured by a bivariate copula can be described through its tail dependence coefficients λ_L and λ_U , which quantify the probability of joint extreme co-movements in the lower and upper tails respectively. For $U_1, U_2 \sim C$, the lower and upper tail dependence coefficients are defined as

$$\lambda_L = \lim_{u \rightarrow 0^+} \frac{C(u, u)}{u}, \quad \lambda_U = \lim_{u \rightarrow 1^-} \frac{1 - 2u + C(u, u)}{1 - u}$$

when the limit exists. Both of these coefficients admit values in $[0, 1]$, with values close to 0 indicating tail independence and values close to 1 indicate strong joint tail behavior. Table A1 shows the tail dependence behavior for the bivariate families described above.

B Datasets

This Section provides supplementary graphical representations of the datasets analyzed in Section 4, which were omitted from the main body due to space constraints. Fig. B1 shows the pairwise scatterplots for every pair of features included in the Wilt dataset. The ordinary and anomalous observations overlap almost perfectly across all variable pairs, suggesting that the anomalies do not arise from extreme marginal behaviors, but from different dependence patterns that exist for ordinary and anomalous observations, as is clearly visible in Fig. 3.

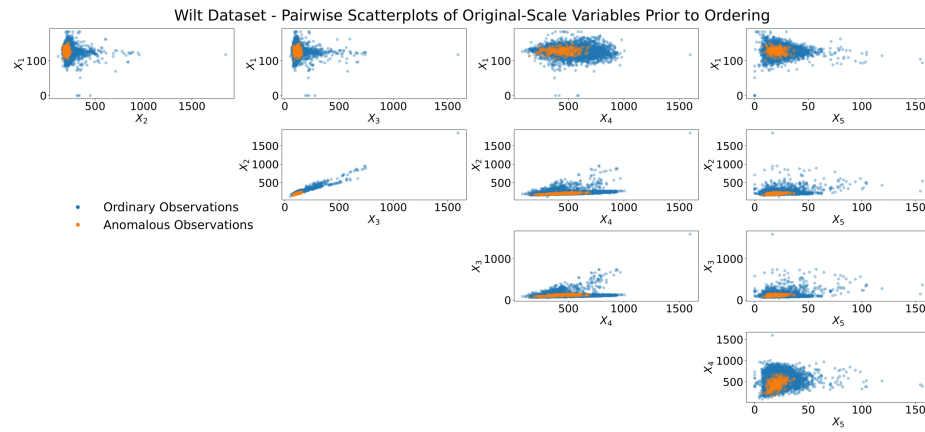


Fig. B1. Wilt: Pairwise scatterplots of the variables in their original scale. Ordinary and anomalous observations are included and coloured based on their label.

Fig. B2 illustrates the multivariate time series from the Sewerage dataset, showing hourly readings for Sensors 1–4. The left panel depicts ordinary conditions, characterized by regular daily patterns and spikes during rainfall events. Right: the anomalous state of Sensor 4, whose readings no longer follow the shared network behavior.

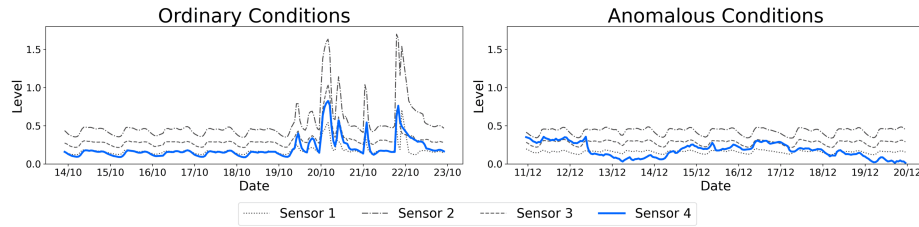


Fig. B2. Sewerage: Hourly level recordings for the four sensors. Left: ordinary behavior, with coherent daily patterns in dry conditions and simultaneous rainfall spikes. Right: the anomalous state of Sensor 4, whose readings no longer follow the shared network behavior.

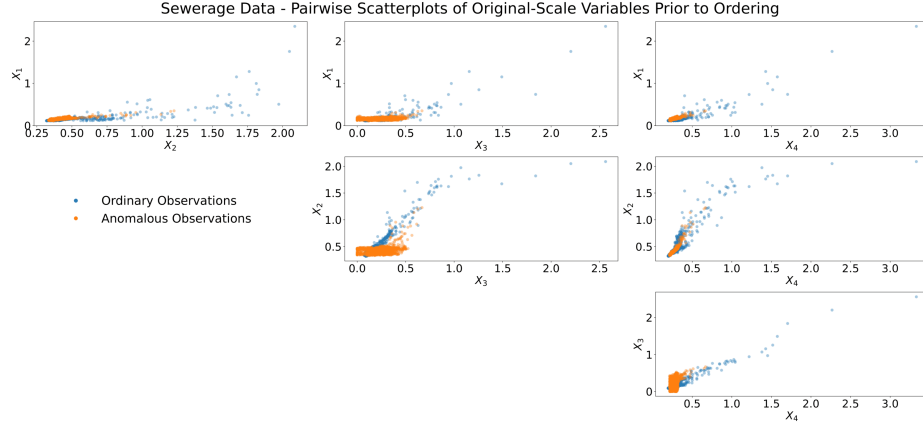


Fig. B3. Sewerage: Pairwise scatterplots of the variables in their original scale. Ordinary and anomalous observations are included and coloured based on their label.

The right panel shows anomalous behavior in Sensor 4 (highlighted in blue), which breaks from the shared dynamics of the network while staying within its operational range.

Fig. B3 shows the pairwise scatterplots for each pair of features included in the real-world Sewerage Dataset. In this scenario the ordinary and anomalous observations overlap significantly for most variable pairs with the exception of those involving feature X_3 , the one containing the reading from the anomalous Sensor 4. This pattern is further confirmed by Fig. B4, where the clearest separations between ordinary and anomalous behavior can be attributed to the same feature. Notably, the second and third panels represent the pseudo-observations which form the inputs of edges (1,2) and (1,3) of the D-vine, whose relevance in identifying anomalies was described in Section 4.

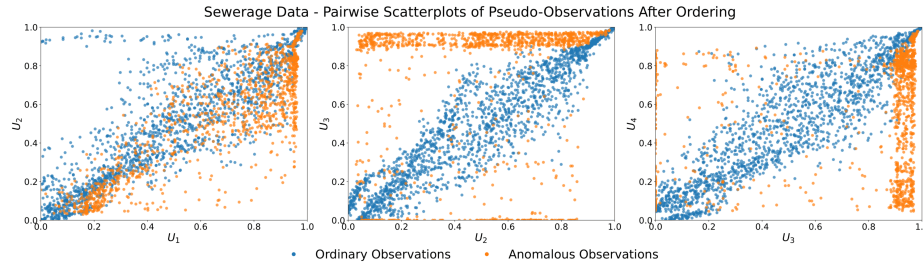


Fig. B4. Sewerage: Pairwise scatterplots of the ordered pseudo-observations. Variables pairs plotted together form the inputs to the pair-copulas at the first tree T_1 .

C Additional Results

This section presents additional experimental results for both datasets analyzed in Section 4, supplementing those reported in the main body.

C.1 Wilt Dataset

Fig. C1 shows the D-vine configuration estimated by the fitting procedure without penalization. Each panel visualizes the joint density induced by the fitted pair-copula combined with standard Gaussian marginals. When compared to Fig. 4, most edges maintain similar family-parameter configurations when fitting with and without penalization, with the exceptions being edges (1,3) and (2,2). Given that these edges are the ones associated with pairs of variables which present a clearer class separation, the penalized version of the fitting procedure is better able to discriminate between the two classes, forcing a configuration which assigns lower density to anomalous observations.

This factor is also visible from Fig. C2, which highlights the standardized edge scores across the calibration and test sets, for both ordinary and anomalous observations. Similarly to Fig. 5, edges (1,3) and (2,2) present higher scores for anomalous observations compared to other edges. However, when comparing with the penalized version, the scores associated to these edges have values lower in magnitude, especially for edge (2,2).

Lastly, Fig. C3 provides a graphical representation showing the classification obtained at these edges.

The results are also presented in numerical form in Table C1.

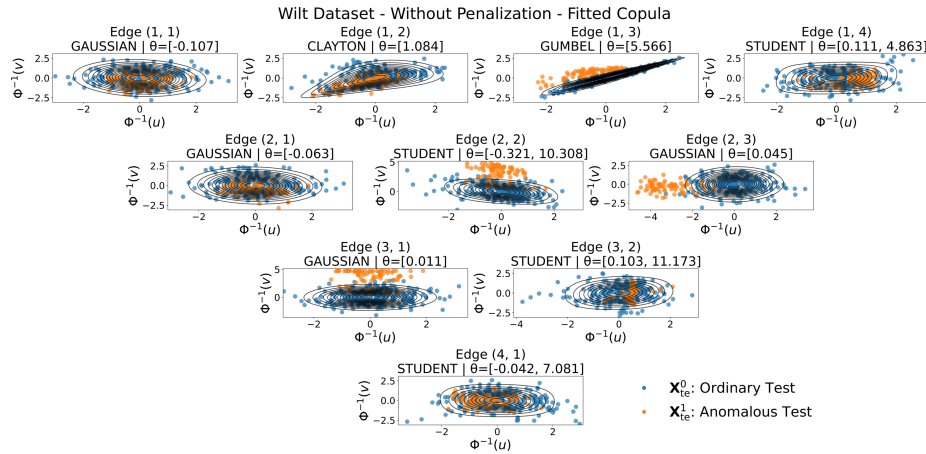


Fig. C1. Wilt: Fitted pair-copulas of the D-vine without penalization, shown as joint densities. Each panel can be read in isolation, as the dependence accumulated in earlier trees is embedded in the inputs of each panel through the h-function cascade.

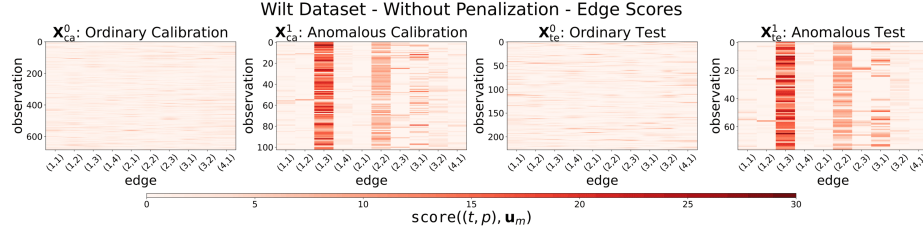


Fig. C2. Wilt: Standardized edge scores $\text{score}((t,p), \mathbf{u}_m)$ for the Wilt dataset across the calibration and test sets, for fitting without penalization.

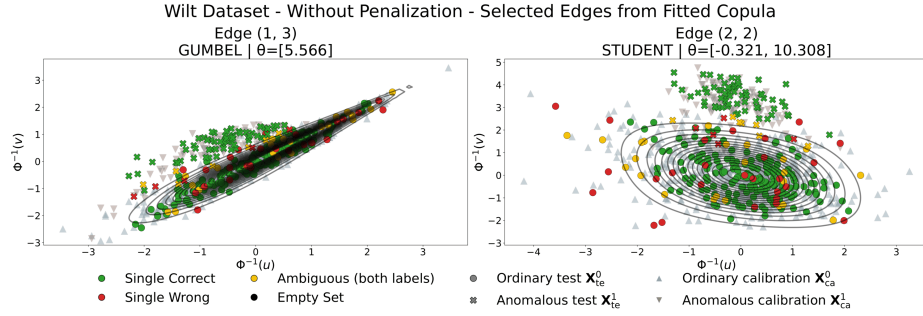


Fig. C3. Wilt: Anomaly detection results under non-penalized fitting at edges (1,3) (left) and (2,2) (right), the two most relevant for anomaly detection. Marker color indicates prediction correctness; shape distinguishes ordinary from anomalous observations.

Dataset	True Class	n	Coverage	Single Correct	Single Wrong	Both Labels	Empty Set
Wilt Dataset	Ordinary	228	0.886	179	26	23	0
	Anomalous	77	0.935	65	5	7	0
Sewerage Data	Ordinary	256	0.887	116	29	111	0
	Anomalous	335	0.925	245	23	67	0

Table C1. Class-conditional conformal prediction results on the Wilt and Sewerage data test sets at $\alpha = 0.1$, D-vine fitted without penalization. Coverage is the fraction of observations whose prediction region contains the true label (target ≥ 0.9). The remaining columns report the composition of the prediction regions.

C.2 Sewerage Data

Similarly to the Wilt dataset, the most notable differences in configurations between the penalized and non penalized versions can be observed for the edges which present a clearer separation between ordinary and anomalous observa-

tions. When comparing Figures C4 and C5, the most notable differences can be observed in edges (1, 2) and (1, 3), which are the edges in T_1 responsible for modeling the features associated with the anomalous Sensor 4.

Fig. C6 shows the standardized edge scores for the fitting procedure without penalization. When comparing with Fig. 7, edges (1, 2), (1, 3) and (3, 1) present higher than average score values but the magnitude is more muted.

Lastly, Fig. C7 provides a graphical representation showing the classification obtained at these edges. Notably, when comparing with Fig. 8, a higher number of observations are assigned both labels as can be seen by the prevalence of yellow-colored observations in all represented edges. This result can also be appreciated from a numerical standpoint in Table C1, where the number of observation for the Sewerage datasets which are assigned both labels is significantly higher than its penalized counterpart in Table 1. For this dataset, the omission of a penalization term manifests with less confident prediction in the overlap regions between the two classes, resulting in more conservative predictions.

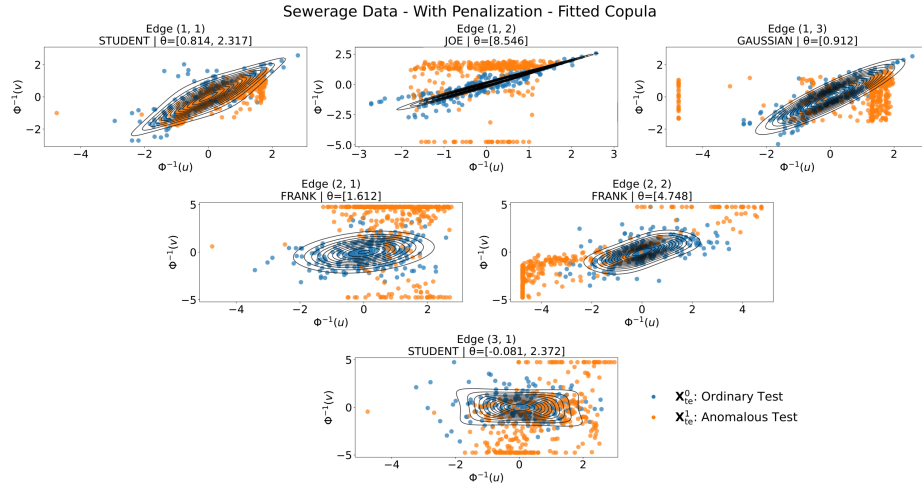


Fig. C4. Sewerage: Fitted pair-copulas of the D-vine with penalization, shown as joint densities. Each panel can be read in isolation, as the dependence accumulated in earlier trees is embedded in the inputs of each panel through the h-function cascade.

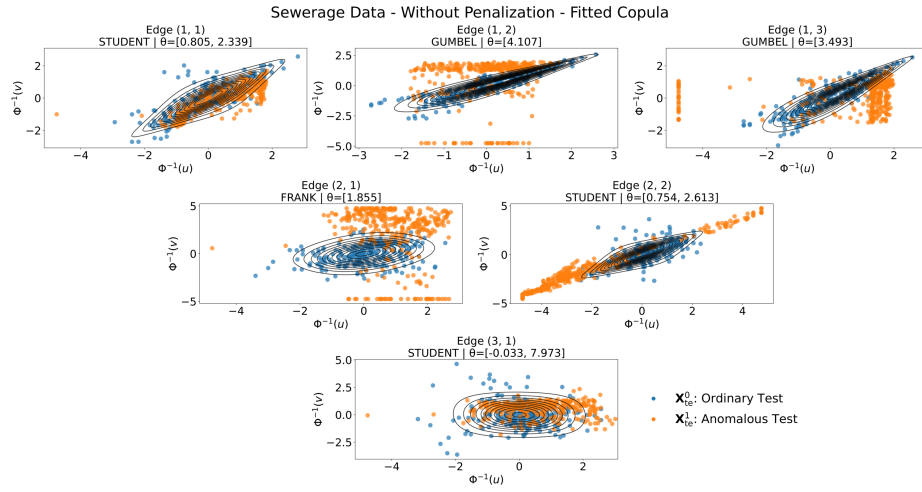


Fig. C5. Sewerage: Fitted pair-copulas of the D-vine without penalization, shown as joint densities. Each panel can be read in isolation, as the dependence accumulated in earlier trees is embedded in the inputs of each panel through the h-function cascade.

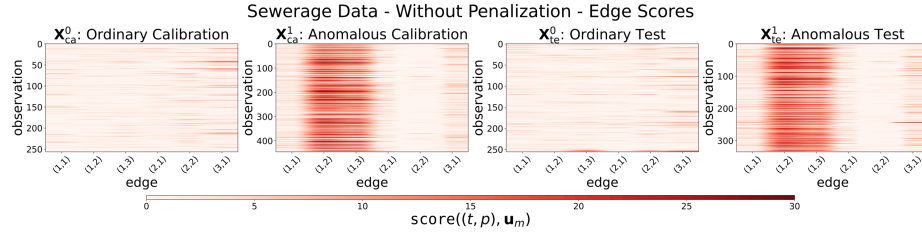


Fig. C6. Sewerage: Standardized edge scores $\text{score}((t, p), \mathbf{u}_m)$ for the Sewerage dataset across the calibration and test sets, for fitting without penalization.

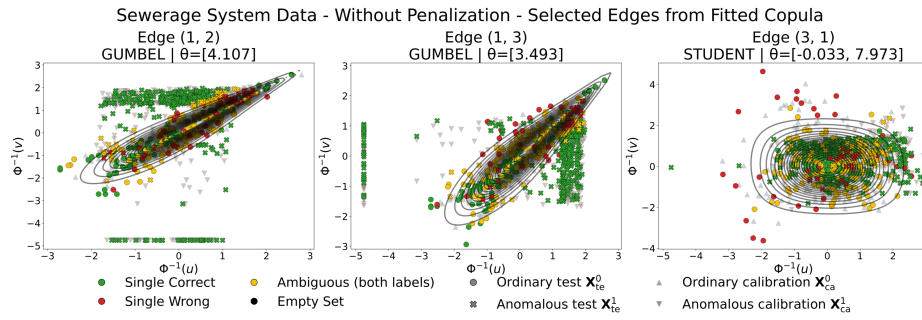


Fig. C7. Sewerage: Anomaly detection results under non-penalized fitting at edges (1, 2), (1, 3) and (3, 1), the three most relevant for anomaly detection. Marker color indicates prediction correctness; shape distinguishes ordinary from anomalous observations.